
VARDHMAN
Airport Solutions

# LED Sequenced Flashing Lighting System

High Intensity, Uni-directional, Inset, Elevated Light

VAS-SFLS-W-L-XX-XX/XX-CAN-02

## Compliance with standards

- ICAO/CAR Annex 14, Volume 1, Doc 9157, Part 4
- FAA E-2628B/ Engg. Brief 67D
- FAA AC-150/5345-51B/AC-150/5345-46F
- IEC TS 61827

## Application

- A series of sequenced flashing lights designed to guide pilots along the correct approach path in low visibility conditions for better conspicuousity by progressively illuminating toward the threshold of the runway.
- High-intensity, Uni-directional, Sequenced Flashing Lighting System used in ICAO category I, II & III runways.

## Specifications

| | |
|---|---|
| Rated Power for Elevated | 38W at 230V |
| Rated Power for Inset | 58.5W at 230V |
| Power Factor | >95% at nominal current |
| Temperature | -40° C to +55° C |
| Life Time of LED | 50,000 hrs |
| Protection Class for Elev. type | IP 65 |
| Protection Class for in-pave. type | IP 67 |
| Protrusion Height (in-pavement) | 6.3mm |
| Color | White |
| Direction | Uni-directional |
| Light fitting Size (in-pavement) | Ø12" |

## Features

- Light distribution and chromaticity conforming to the specifications as per ICAO Annex 14 Volume 1 standard.
- Each lighting fixture has its separate unit control cabinet so that it has concise cable layout, which is easier for maintenance.
- Both main control cabinet and unit control cabinet have self-contained CPUs, which runs independently and operate in synergy via bus communication.
- The system is equipped with online faults detection function for easy access to real time information.
- Main control cabinet equipped with LCD display, which accumulates records of system operation.
- Monitoring system has remote control operation and updates of operational status.
- Circuits, power and communication cables are equipped with lightning protection, conforming to the requirements of standards.
- The elevated fixture can be connected to both one- or two-inch frangible tube, which ensures convenient and secure installation and forged frangible couplings with precision machining, which complies with standards requirement.
- The optical prism adopts special tempered glass to prevent the prism from bursting due to drastic changes in temperature.


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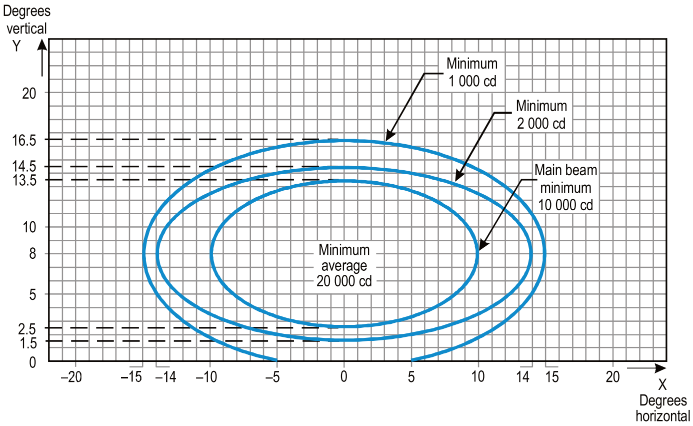
## Features

- Valve for watertightness test.
- The main body of the light fitting is made of aluminium alloy which is strong and has good anti corrosion performance.
- High precision machining to ensure all dimensional quality and precision is achieved as per design specification.
- Elevated type and inset type version.
- Style L-823 connectors are used.

## Photometric Data




## Accessories

| No. | Description | Order number |
|---|---|---|
| 1 | Main Control Box Panel | VAS-66110 |
| 2 | Flashing Light | VAS-66230 |
| 3 | Unit Control Box | VAS-66220 |
| 4 | Unit Control Box circuit board | VAS-79108 |
| 5 | Light circuit board | VAS-79107 |
| 6 | Reflector | VAS-31147 |
| 7 | Sealing ring | VAS-41119 |
| 8 | Rear cover seal | VAS-41111 |

| Approach Light ICAO Fig. A2-1 | |
|---|---|
| Beam angle | Horizontal ±15°, vertical ±5° |
| Flash frequency | once or twice per second |
| Light intensity | I class 2% (150 ~ 450 cd) |
| | II class 10% (800 ~ 2,000 cd) |
| | III class 100% (8,000 ~ 20,000 cd |
| Reflector ratio | > 95% |
| Elevation angle | 0 to 15° vertical |
| Chromaticity | ICAO Annex. 14 Fig A1-1b |

### Part Code

| Part Code Information | |
|---|---|
| Elevated SFLS LED | VAS-ESFLS-W-L |
| Inset SFLS LED | VAS-ISFLS-W-L |
| Master Control Unit | VAS-MCU |
| Flashing Control Unit | VAS-FCU |

## Ordering information

VAS-SFLS-W-L-XX-XX/XX-XXX-XX

XX = Flash frequency
- 01 : 1 Hz
- 02 : 2Hz

XXX = Communication Mode
- CAN = CAN communication
- PLC = Carrier communication

XX/XX = Number of Heads
- 00/01 = 0 Elevated light/1 Inset light
- 01/01 = 1 Elevated light/1 Inset light
- 01/09 = 1 Elevated light/9 Inset light
- 19/02 = 19 Elevated light/2 Inset light
- 30/00 = 30 Elevated light/0 Inset light

XX = Power supply
- 3P =Three-phase power supply
- 1P = Single-phase power supply

L= LED

W= White

SFLS = Sequenced flashing lighting stystem

VAS= VARDHMAN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## Master Control Unit


### Power Supply

- Connected to AC 220V main power supply.
- The power switch in the cabinet, along with its supporting circuits, controls the system power ON/OFF.

### Control Mode

The Local/Remote Control selection switch on the main control cabinet panel allows switching the control mode of the sequential flash system between:

- Local Control
- Tower Control

### Operation Interface

A color touch screen interface provides the following functions:

- Start and stop of the sequential flash lights.
- Adjustment of flash light intensity levels.
- Switching of flash frequency.
- Display of system status information.

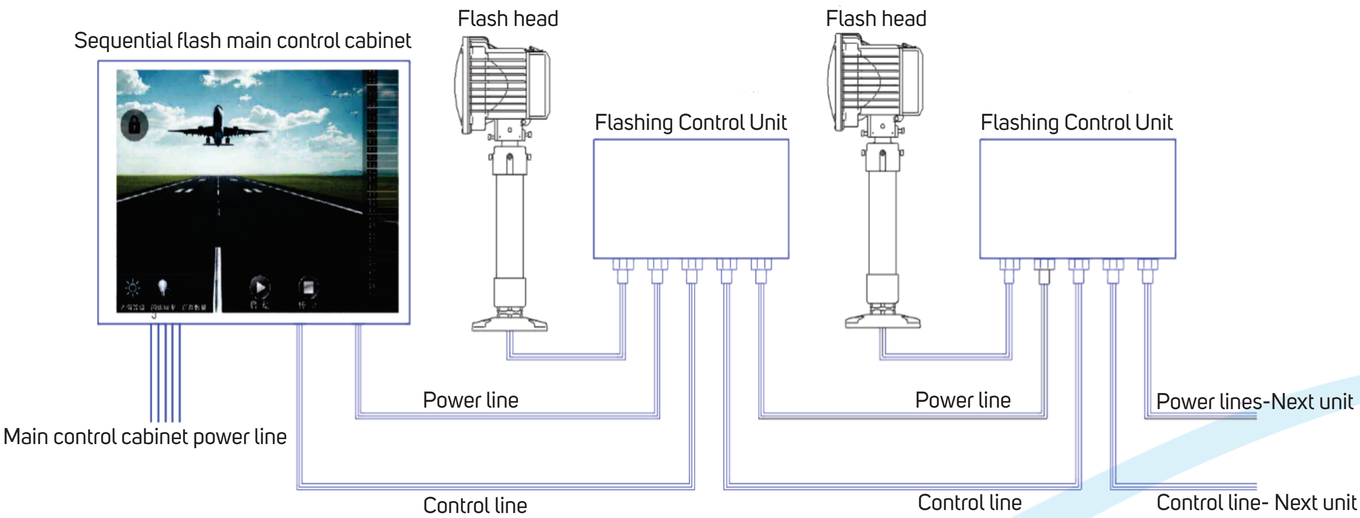
### Status Indication

- The working status of each flash unit is indicated by an LED board on the panel.

### Connections

- Power lines and communication lines for the sequential flash units are drawn from the main control cabinet.
- These lines are connected in sequence to the control box of each unit.

## Overall connection diagram:

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# Flashing Control Unit

### Connections

The unit control box is connected to:

- The main control cabinet of the system, or
- The power line and communication line output from the previous unit.
- It provides power line and communication line outputs to the next unit in the sequence.

### Control and Functions

The sub-control board and its associated circuits inside the unit control box perform the following functions:

- Three-level light intensity control of the flash.
- Flash triggering and flash detection.
- Storage of flash-related data.
- Feedback and return of status information to the main control cabinet.

### Safety and Protection

Equipped with a door lock switch:

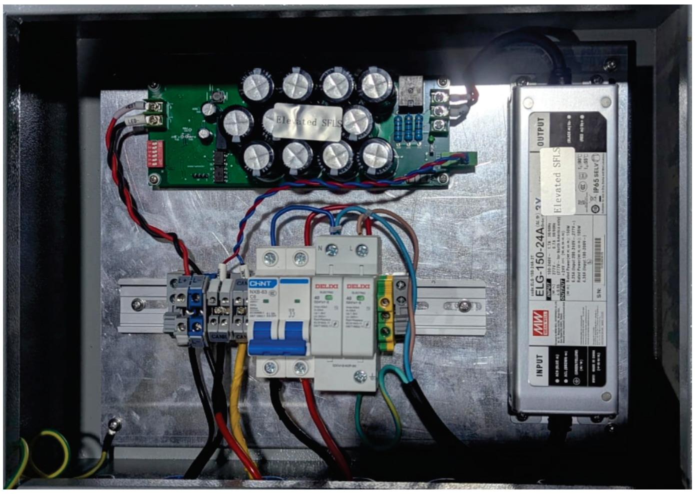
- When the box is opened for maintenance, the power is automatically cut off.
- This does not affect the normal operation of other unit control boxes in the system.

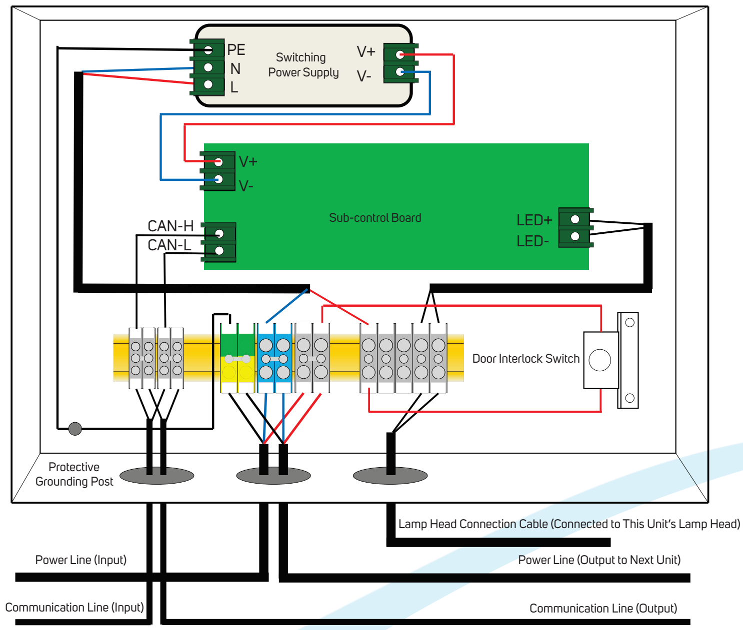
### Environmental Protection

An anti-condensation respirator device is installed at the bottom of the box to:

- Reduce humidity inside the control box.
- Prevent condensation and maintain reliable operation.

## Overall connection diagram:

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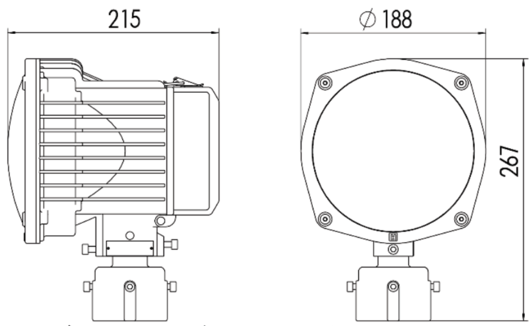
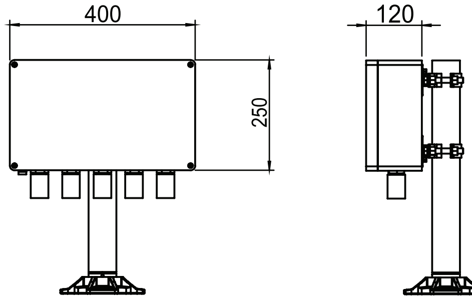
## Dimensions

SFL Elevated type




(Front View) (Side View)

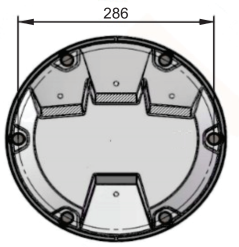
Light feature



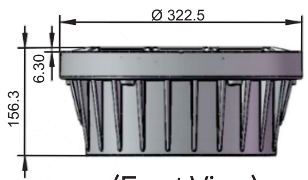

(Front View) (Side View)

Flashing Control unit box

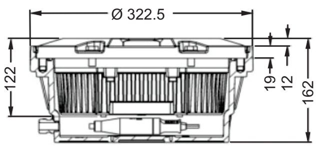
SFL Inset type




(Top View)

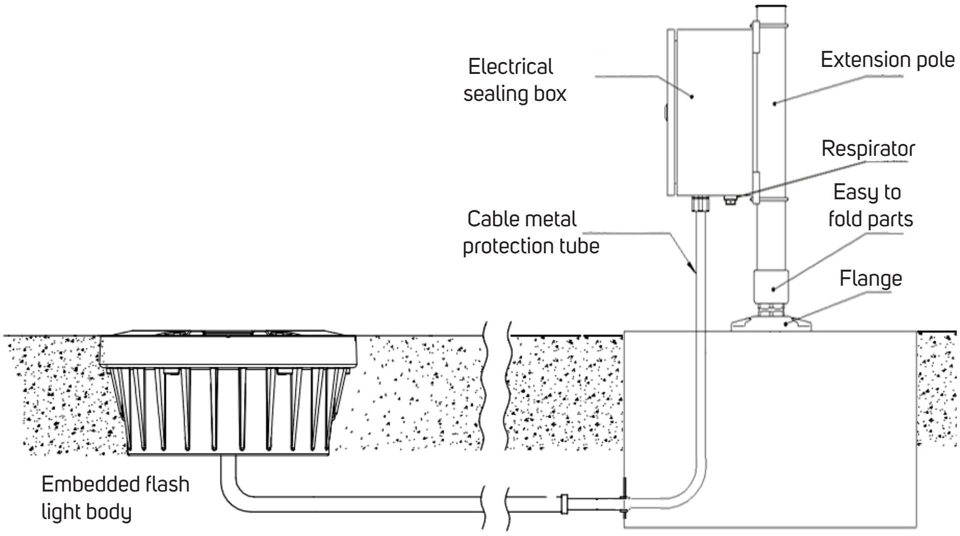



(Front View)

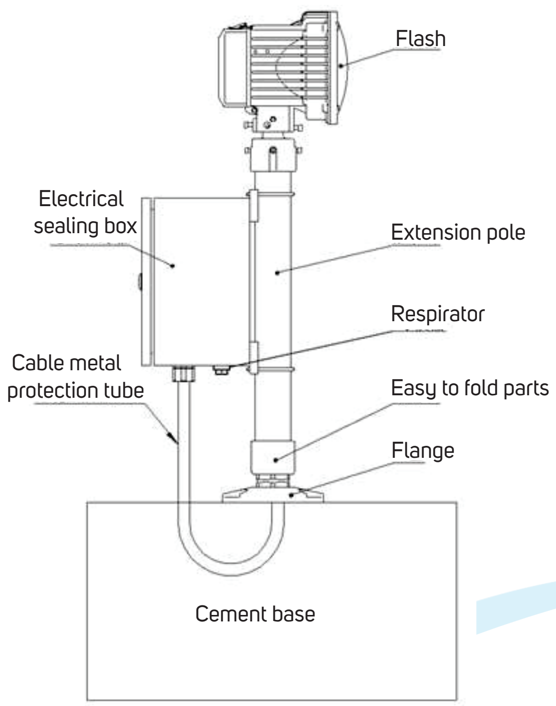



(Section View)

## Installation




Flush Mount




Vertical installation

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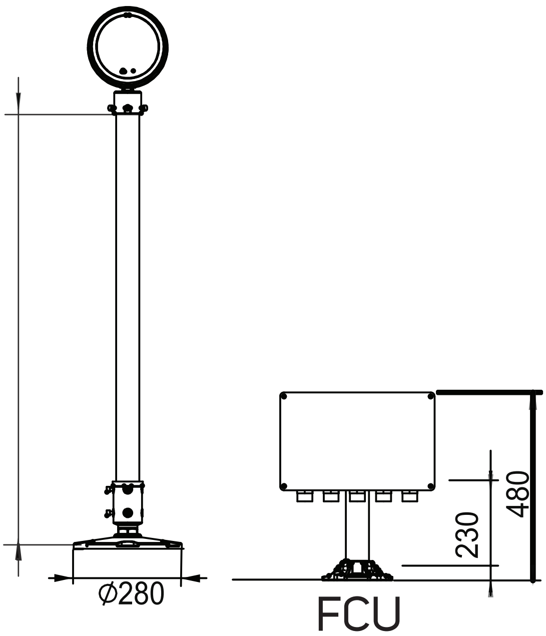


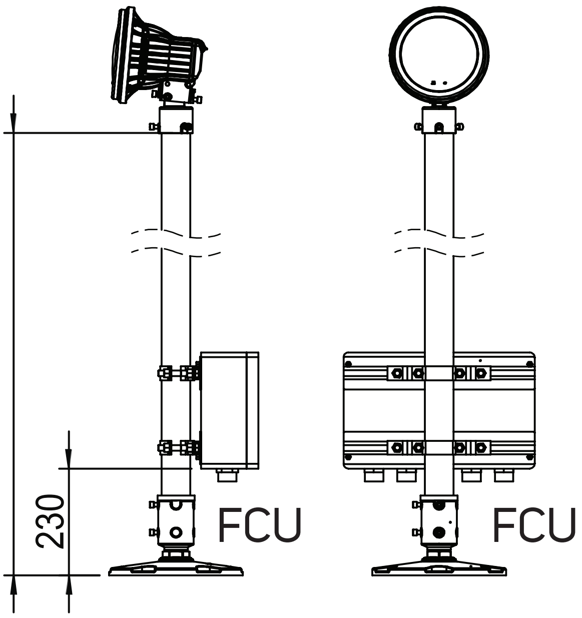


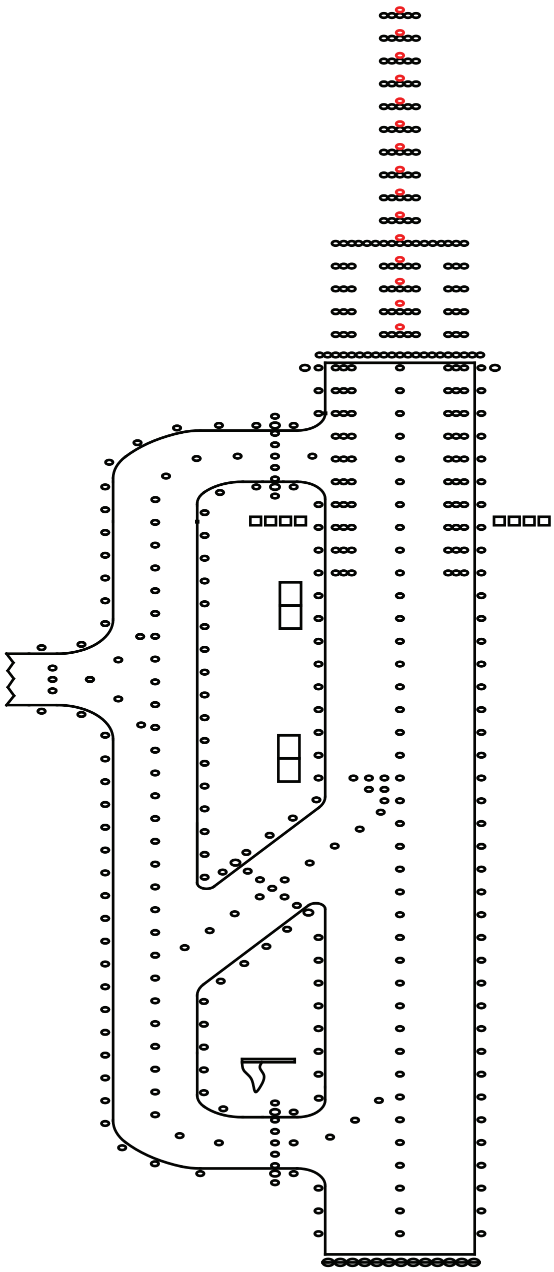

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